

# *Porpolomopsis calyptriformis*, Pink Waxcap

## Errata version

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## Taxonomy

| Kingdom | Phylum        | Class          | Order      | Family         |
|---------|---------------|----------------|------------|----------------|
| Fungi   | Basidiomycota | Agaricomycetes | Agaricales | Hygrophoraceae |

**Taxon Name:** *Porpolomopsis calyptriformis* (Berk.) Bresinsky

### Synonym(s):

- *Agaricus calyptriformis* Berk.
- *Hygrocybe calyptriformis* (Berk.) Fayod
- *Porpolomopsis calyptriformis* var. *calyptriformis* (Berk.) Bresinsky

### Common Name(s):

- English: Pink Waxcap
- French: Hygrophore en Capuchon

### Taxonomic Source(s):

Index Fungorum Partnership. 2018. Index Fungorum. Available at: <http://www.indexfungorum.org>.

### Taxonomic Notes:

Sequencing data from records assigned to *em style="margin: 0px; padding: 0px; font-family: Arial, sans-serif; font-size: 11px; background-color: rgb(255, 255, 255);">Porpolomopsis calyptriformis* in the USA (from the Great Smoky Mountain National Park) show that they are not the same species (Lodge *em style="margin: 0px; padding: 0px; font-family: Arial, sans-serif; font-size: 11px; background-color: rgb(255, 255, 255);">et al.* 2014, Lodge *em style="margin: 0px; padding: 0px; font-family: Arial, sans-serif; font-size: 11px; background-color: rgb(255, 255, 255);">et al.* unpublished results). Unsequenced collections in Mushroom Observer from the USA (Ohio, California and North Carolina), and from Canada (Quebec), don't fit the typical morphology of the European *em style="margin: 0px; padding: 0px; font-family: Arial, sans-serif; font-size: 11px; background-color: rgb(255, 255, 255);">P. calyptriformis* and also probably aren't the same species (Lodge, personal communication). However, because a variety of *em style="margin: 0px; padding: 0px; font-family: Arial, sans-serif; font-size: 11px; background-color: rgb(255, 255, 255);">P. calyptriformis* has been described from the Dominican Republic at high elevation on the island of Hispaniola, one can't rule out the possibility that it occurs in the western hemisphere (Lodge, personal communication).

## Assessment Information

**Red List Category & Criteria:** Vulnerable A2c+3c+4c [ver 3.1](#)

**Year Published:** 2019

**Date Assessed:** November 1, 2018

### Justification:

*Porpolomopsis calyptriformis* is only known from Europe and has shown a considerable decline in many countries. It is so far the only representative of the *Porpolomopsis* genus in the diverse fungal communities mainly associated with semi-natural European grasslands, the so-called wax cap

grasslands. These habitats, formerly characterized by extensive management systems, have undergone a major decline in area in recent decades. Major threats relate to changes in management regime (either intensification or abandonment), airborne nitrogen deposition, and climate change. It is assessed as Vulnerable (VU) under criteria A2c+3c+4c, because of past, ongoing, and estimated future habitat decline between 30-50% in a 50-year (3 generation length) period.

## Geographic Range

### Range Description:

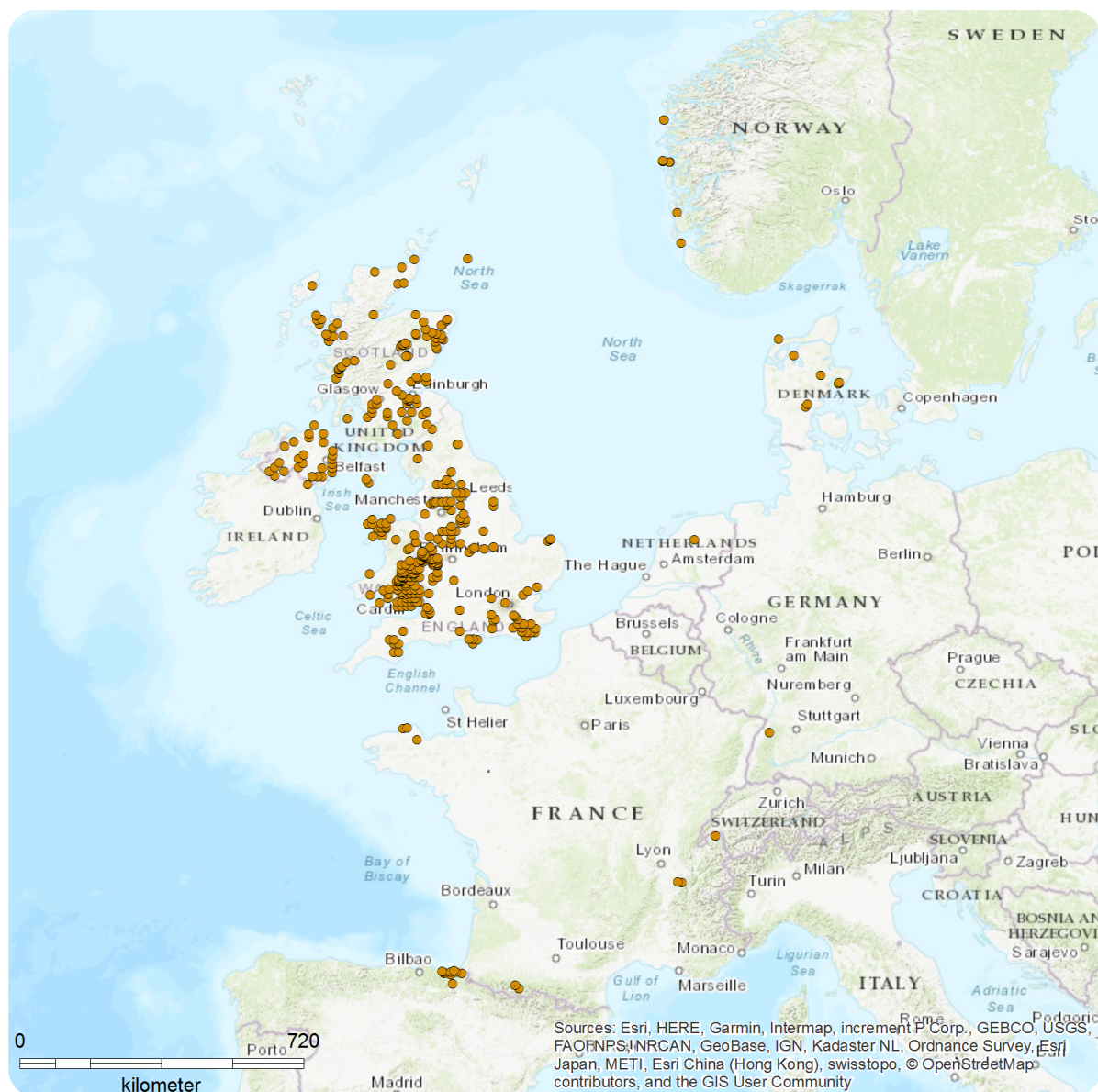
The species is so far only known from Europe. It is reported from several countries (but often from single locations) with a stronghold in the UK. In Scandinavia, the species is restricted to Denmark and three locations on the southern coast of Norway. The eastern limits of distribution are unclear due to lack of data (e.g. an isolated record in Ukraine).

### Country Occurrence:

**Native:** Austria; Belgium; Croatia; Czechia; Denmark; Estonia; Faroe Islands; France; Germany; Hungary; Ireland; Italy; Latvia; Liechtenstein; Lithuania; Netherlands; Norway; Portugal; Romania; Russian Federation; Serbia; Slovakia; Slovenia; Spain; Switzerland; Ukraine; United Kingdom

# Distribution Map

*Porpolomopsis calyptriformis*



Range

- Extant (resident)

Compiled by:

IUCN



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

## Population

The population size probably exceeds 20,000 mature individuals but exact figures are unknown. The population is decreasing in all known countries of occurrence, caused by a cessation of small-scale farming and traditional methods of grassland management.

The mid-term review of the EU biodiversity strategy towards to 2020 (European Commission, 2015) identifies “an increase in habitat change” in grassland ecosystems as the future trend, with high impact on biodiversity. The report concludes that although some progress was made, this has not yet halted the trend of degradation of ecosystems and services, including grasslands: “Natural grasslands are still being turned into arable land and built-up areas; extensive agricultural land is being converted into forms of more intensive agriculture and, mainly as a result of abandonment, into forest.” According to another reference document, the “State of Nature in the EU” (European Environmental Agency, 2015), natural and semi-natural grasslands have undergone a major decline in recent decades. About 49% of EU assessments for the 45 grassland habitat types of community interest are “unfavourable-bad”. Moreover, almost 50% of grassland-related birds are declining and the conservation status of other species is mostly “unfavourable”. Grassland butterflies, for example, are declining severely and there is no sign of levelling off. Accordingly, the Red List of European Habitats (Janssen *et al.* 2017) reports that just about half (53%) of the grassland habitats in Europe are threatened to some degree.

NBN Atlas - 1300 records. All evaluations of grassland habitat types in the UK have an “unfavourable-bad” conservation status (European Environmental Agency, 2015). Atlas of Danish Fungi - 15 records. Mushroom Observer - Less than 20 records mainly in the USA (likely another species). GBIF - 23 records under *P. calyptriformis* (Denmark, 15; UK, 4; USA, 4- likely another species), 1257 records under *H. calyptriformis* var. *calyptriformis* (UK, 122; Slovenia, 23; Ireland, 9; USA, 4- likely another species), 395 under *H. calyptriformis*, (UK, 351; USA, 12- likely another species; Norway, 11; Austria, 4; Canada, 4 - likely another species; France, 4; Denmark, 4; Australia 1- almost certainly a misidentification, confusion with sister species *P. lewellinae*; Germany, 1; Spain 1).

**Current Population Trend:** Decreasing

## Habitat and Ecology (see Appendix for additional information)

*Porpolomopsis calyptriformis* is mainly associated with natural or semi-natural grasslands, which are among the most threatened habitats in Europe. It is terrestrial with unknown nutrition mode, but likely biotrophic, possibly associated with non-woody plants (Seitzman *et al.* 2011, Halbwachs *et al.* 2013, Lodge *et al.* 2014). Typical habitats include meadows, grazed pastures, parks, lawns, churchyards and cemeteries. Lowland and montane. The species occurs mainly in the following Natura 2000 grassland habitats: 6150 - Siliceous alpine and boreal grasslands, 6170 - Alpine and subalpine calcareous grasslands, 6210 - Semi-natural dry grasslands and scrubland facies on calcareous substrates (Festuco-Brometalia); the most frequent habitat, 6230 - Species-rich *Nardus* grasslands on siliceous substrates in mountain areas (and sub-mountain areas in Continental Europe), 62A0 - Eastern sub-Mediterranean dry grasslands, 6410 - Calcareous fens with *Carex lasiocarpa*, 6510 - Lowland hay meadows (*Alopecurus pratensis*, *Sanguisorba officinalis*), 6520 - Mountain hay meadows.

**Systems:** Terrestrial

## Threats (see Appendix for additional information)

The species is threatened by the loss of its habitat. Natural and semi-natural grasslands, formerly characterized by extensive management systems, have undergone a major decline in recent decades. A major threat relates to the conversion of grasslands into more productive forms of intensive stock management, reported to involve liberal use of chemical fertilizers rather than the traditional dung. Such nutrient input can be also increased by air pollution in the form of nitrogen deposition. A second major threat is the abandonment of traditional management. Generally, this is due to the abandonment of stock management or, where cutting for hay has been traditional, lack of the necessary mowing regime. This results in a reversion to scrub and woodland. In the opposite direction, there is also sometimes shifts towards systems of intensive crop cultivation, with use of pesticides and/or herbicides. Either way, changes in vegetation cover are detrimental to the fungal community. Such changes are often part of wider demographic, socio-economic and cultural shifts across large parts of the European rural landscape. In Alpine and sub-alpine grasslands there is also a concern that they may be strongly impacted by climate change, for example through milder winters with reduced snow-lie and longer growing seasons.

## **Conservation Actions (see Appendix for additional information)**

Both site protection and management of habitats are important conservation actions. Appropriate management plans of semi-natural grasslands (including grazing/mowing) are needed to halt the frequent decrease of habitat quality even if sites are within the Natura 2000 network. According to the “State of Nature in the EU” report (European Environmental Agency, 2015), only 50 % of sites within the network were reported as having comprehensive management plans by end 2012 and much stronger conservation efforts are needed. Restoration plans may depend on effective incentives for re-establishing low-input arable agriculture and traditional stock management. Outside of Natura 2000 sites, preservation of appropriate habitats for grassland fungi, such as *P. calyptriformis*, will require consideration of the most suitable approach to ensure no net loss of biodiversity and ecosystem services, e.g. policies targeting a decrease in nitrogen and phosphorus inputs in agriculture.

## **Credits**

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## Appendix

### Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Habitat                                    | Season | Suitability | Major Importance? |
|--------------------------------------------|--------|-------------|-------------------|
| 4. Grassland -> 4.4. Grassland - Temperate | -      | Suitable    | -                 |

### Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Threat                                                                                                                 | Timing  | Scope | Severity | Impact Score |
|------------------------------------------------------------------------------------------------------------------------|---------|-------|----------|--------------|
| 1. Residential & commercial development -> 1.1. Housing & urban areas                                                  | Ongoing | -     | -        | -            |
| 11. Climate change & severe weather -> 11.1. Habitat shifting & alteration                                             | Ongoing | -     | -        | -            |
| 2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.1. Shifting agriculture                | Ongoing | -     | -        | -            |
| 2. Agriculture & aquaculture -> 2.1. Annual & perennial non-timber crops -> 2.1.3. Agro-industry farming               | Ongoing | -     | -        | -            |
| 2. Agriculture & aquaculture -> 2.3. Livestock farming & ranching -> 2.3.3. Agro-industry grazing, ranching or farming | Ongoing | -     | -        | -            |
| 9. Pollution -> 9.3. Agricultural & forestry effluents -> 9.3.1. Nutrient loads                                        | Ongoing | -     | -        | -            |
| 9. Pollution -> 9.3. Agricultural & forestry effluents -> 9.3.3. Herbicides and pesticides                             | Ongoing | -     | -        | -            |

### Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

| Conservation Actions Needed                                            |
|------------------------------------------------------------------------|
| 1. Land/water protection -> 1.1. Site/area protection                  |
| 1. Land/water protection -> 1.2. Resource & habitat protection         |
| 2. Land/water management -> 2.1. Site/area management                  |
| 2. Land/water management -> 2.3. Habitat & natural process restoration |
| 5. Law & policy -> 5.1. Legislation -> 5.1.1. International level      |

|                                                                          |
|--------------------------------------------------------------------------|
| <b>Conservation Actions Needed</b>                                       |
| 5. Law & policy -> 5.1. Legislation -> 5.1.2. National level             |
| 6. Livelihood, economic & other incentives -> 6.4. Conservation payments |

## Additional Data Fields

|                               |
|-------------------------------|
| <b>Habitats and Ecology</b>   |
| Generation Length (years): 17 |

## Errata

**Errata reason:** This is an errata assessment to correct the map - an incorrect map showing a small part of its distribution as the global distribution had previously been published.

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